

Maryam Hejri

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RESEARCH EXPERIENCE

Biophysical constraints on bacterial biofilms growth and survival Jan 2023- present

Graduate Researcher, Quantitative Biosciences Program | Georgia Institute of Technology, Atlanta, Georgia

- Investigated biophysical mechanisms underlying biofilm growth arrest and survival under nutrient limitation and desiccation using *Vibrio cholerae* and *Aeromonas veronii* strains
- Designed and conducted microbiology and imaging experiments, including white-light interferometry, fluorescence microscopy, confocal microscopy, and viability assays (CFU counting)
- Analyzed biofilm morphology, biomass, and growth dynamics using Python-based data analysis and statistical methods

Research in COVID-19 epidemiological modeling April 2021-Nov 2021

Undergraduate researcher, Department of Mathematics | Amirkabir University of Technology, Tehran, Iran

- Built a SIQRD epidemiological differential equation-based model using multi-source COVID-19 data and Monte Carlo parameter estimation to predict pandemic waves and evaluate vaccination and quarantine strategies

Topological effects and statistical physics of copolymers August 2020- July 2021

Undergraduate Researcher, Department of Physics | Amirkabir University of Technology, Tehran, Iran

- Studied folding mechanisms and self-assembly of biomolecules and assessed topological dynamics on the functionality and toxicity of biopolymers such as DNA and proteins

PUBLICATIONS

- A. Krueger, B. Bogati, R. Copeland, **M. Hejri**, J. Zhu, D. S. Weiss, P. J. Yunker. “Rapid Bacterial Identification through Biophysical Surface Topography Features” bioRxiv 2026.03.12.711416; doi: <https://doi.org/10.64898/2026.03.12.711416>
- H. Yoon, K. Yoon, K. Tadesse, **M. Hejri**, R. Koorse, H. Yoo, H. Katani, W. C. Ratcliff, P. J. Yunker, S. Jia. “Real-time Acquisition-to-display Panoramic Super-resolution Microscopy.” SSRN 6915098 (2026)
- A.R. Pokhrel*, R. Copeland*, **M. Hejri**, T. E. R. Belpaire, G. Steinbach, S. L. Ng, B. K. Hammer, P. J. Yunker. “Cell-substrate friction controls biofilm development.” bioRxiv (2025). doi.org/10.1101/2025.07.11.664457

EDUCATION

Georgia Institute of Technology | Atlanta, GA August 2022 – present

Ph.D. Candidate, Quantitative Biosciences (GPA: 3.75/4.00)

Amirkabir University of Technology | Tehran, Iran

Sept 2017 – Sept 2021

Bachelor of Science, Department of Physics (GPA: 18.04 / 20)

SKILLS

Data Science & Quantitative Analysis: statistical analysis, mathematical modeling, data analysis, image analysis, predictive modeling

Programming & Software: Python (Pandas, NumPy, SciPy, Matplotlib, h5py, scikit-learn), ImageJ/Fiji, Git, GitHub Copilot

Imaging & Advanced Microscopy: white-light interferometry, confocal laser scanning microscopy (CLSM), fluorescence microscopy, time-lapse imaging, topographical characterization

Experimental Bioassays: biofilm assay development, cell culture & inoculation, viability assays, desiccation & starvation assays

Soft Skills & Collaboration: interdisciplinary collaboration, technical writing, public speaking, journal reviews

TALKS AND PRESENTATIONS

APS Global Physics Summit, In-person talk on “What limits biofilm growth” | March 2026, Denver, Colorado

APS Global Physics Summit, In-person talk on “Biofilm dynamics under starvation” | March 2025, Anaheim, California

American Physical Society Conference, In-person talk on “Biophysical constraints on vertical biofilm growth” | March 2024, Minneapolis, Minnesota

TEACHING AND MENTORING

Machine Learning of Viral Molecular Dynamics Workshop¹, Co-organizer and instructor | May 2023, Georgia Tech, Atlanta, Georgia

Undergraduate Research Mentor | Yunker Lab, School of Physics, Georgia Institute of Technology | Spring 2026 – Summer 2026

Mentored two undergraduate researchers in biofilm experimental design and quantitative image analysis, training them in wet lab, Interferometry imaging, and Python-based image analysis

Teaching Assistant, Introductory Physics Laboratory | Georgia Institute of Technology — Fall 2022, Fall 2025

Led weekly laboratory sections for ~30 students; supervised experimental procedure and graded lab reports

Journal Reviewer | Physical Review X |2024, Cell System |2024, Biophysical Journal D |2025